

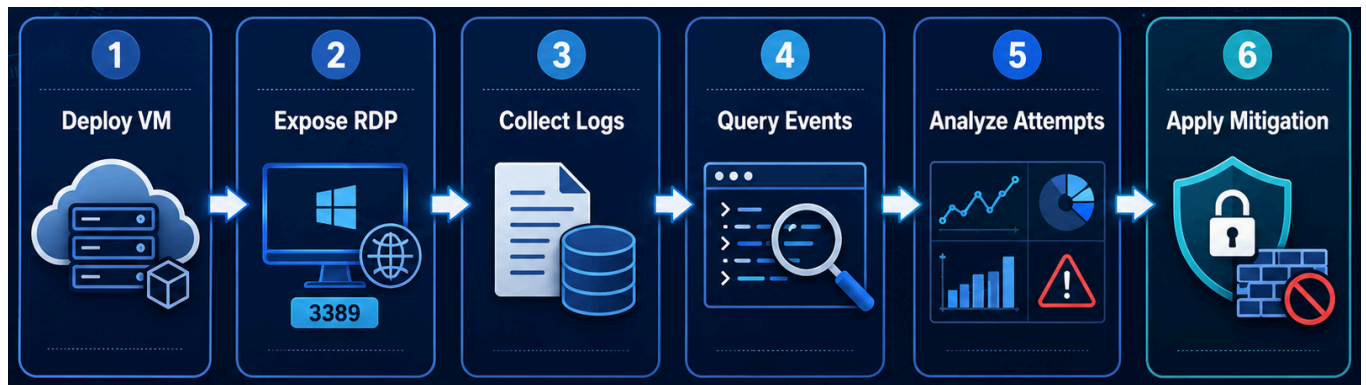
Azure RDP Honeypot Investigation Report

1. Project Overview

This project documents the deployment and monitoring of a controlled Azure-based honeypot environment designed to observe unauthorized Remote Desktop Protocol (RDP) login attempts against a publicly exposed virtual machine.

The main objective of the lab was to understand how quickly an internet-facing Windows machine can attract authentication attempts, how these events appear in Azure Log Analytics, and how basic investigation techniques can be used to identify suspicious login activity.

The project followed a basic security operations workflow:



2. Lab Objective

The objective of this lab was to:

- deploy a Windows virtual machine in Microsoft Azure,
 - expose RDP access temporarily to the public internet,
 - collect Windows event logs using Azure Log Analytics,
 - identify failed RDP authentication attempts,
 - analyze the volume and characteristics of login attempts,
 - verify whether any unauthorized successful logons occurred,
 - document mitigation steps to reduce exposure.
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3. Lab Architecture

The lab environment consisted of the following Azure resources:

Component	Description
Virtual Network	Isolated Azure network used to host the virtual machine
Virtual Machine	Windows-based VM exposed temporarily over RDP
Network Security Group	Controlled inbound access to the VM
Public IP Address	Allowed external access to the VM
Log Analytics Workspace	Collected and stored Windows event logs
Azure Monitor Agent / Data Collection Rule	Forwarded VM event logs to Log Analytics

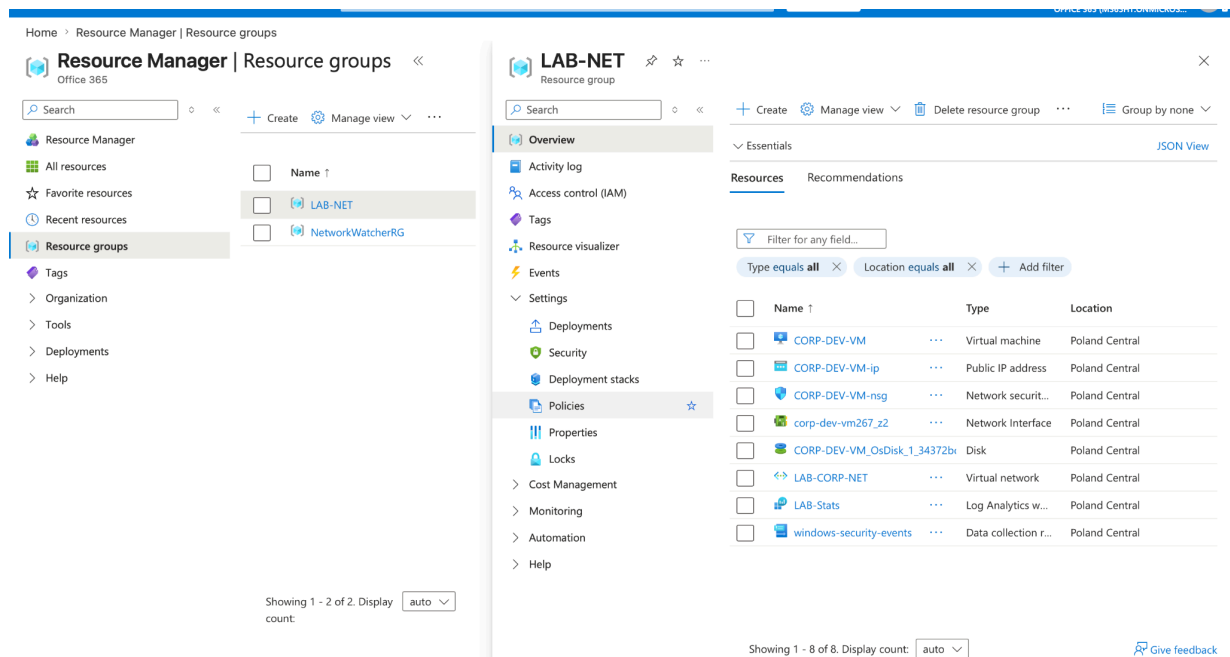


Figure 1. Resource group with Virtual Network, VM and Disc

Architecture Diagram

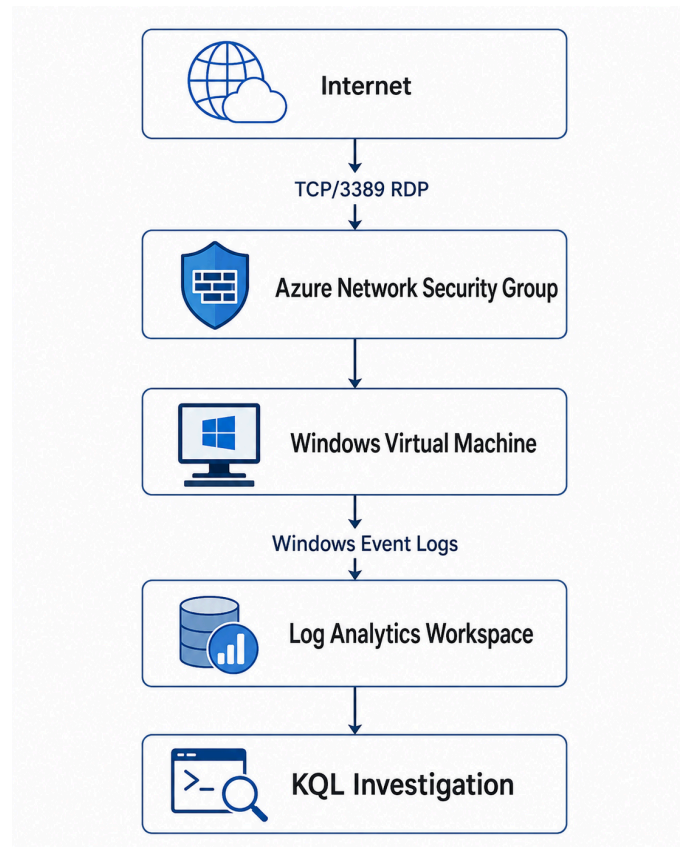


Figure 2. Architecture Diagram

4. Scope of the Investigation

The scope of the lab was limited to observing authentication activity against the exposed RDP service.

The investigation focused on:

- failed logon attempts,
- successful logon events,
- source IP addresses,
- attempted usernames,
- authentication activity over time,
- mitigation through Network Security Group hardening.

This project did not involve malware analysis, exploitation, offensive scanning, or interaction with attacking infrastructure.

5. Security Configuration

For the observation period, the VM was intentionally exposed over RDP.

Initial Exposure Rule

Setting	Value
Protocol	TCP
Port	3389
Source	Any
Action	Allow
Purpose	Controlled RDP honeypot exposure

This configuration was used only for a limited observation window.

6. Log Collection

Windows event logs were collected into Azure Log Analytics. The collected logs were queried using Kusto Query Language (KQL).

The investigation primarily focused on Windows logon events:

Event ID	Description
4625	Failed logon attempt
4624	Successful logon

7. KQL Queries Used

7.1 Failed RDP Logon Attempts

```
Event
| where EventID == 4625
| project TimeGenerated, Computer, EventID, RenderedDescription
| order by TimeGenerated desc
```

7.2 Successful Logon Review

```
Event
| where EventID == 4624
| project TimeGenerated, Computer, EventID, RenderedDescription
| order by TimeGenerated desc
```

7.3 Failed Logons Over Time

```
Event
| where EventID == 4625
| summarize FailedAttempts=count() by bin(TimeGenerated, 15m)
| order by TimeGenerated asc
```

7.4 Extract Source IP Addresses

```
Event
| where EventID == 4625
| extend SourceIP = extract(@"Source Network Address:\s+([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)", 1, RenderedDescription)
| where isnotempty(SourceIP) and SourceIP != "-"
| summarize FailedAttempts=count() by SourceIP
| order by FailedAttempts desc
```

7.5 Attempted Usernames

```
Event
| where EventID == 4625
| extend AccountName = extract(@"Account Name:\s+([^\r\n]+)", 1, RenderedDescription)
| where isnotempty(AccountName)
| summarize Attempts=count() by AccountName
| order by Attempts desc
```

8. Key Findings

During the controlled observation period, the VM received a high number of failed RDP login attempts.

Summary of Observed Activity

Metric	Result
Observation window	Approximately 3 hours
Failed RDP login attempts	Approximately 30,000
Successful suspicious logons	None identified

Exposed service RDP / TCP 3389

Primary event analyzed Event ID 4625

The volume of failed logon attempts indicates that the public RDP service was rapidly discovered by automated scanning or brute-force infrastructure.

9. Interpretation of Findings

The observed activity strongly suggests automated credential-guessing or brute-force attempts against the exposed RDP service.

The VM did not need to be advertised or manually targeted. Once the RDP port was exposed to the public internet, repeated authentication attempts began to appear in the logs.

This demonstrates a key security lesson:

Publicly exposed administrative services can attract automated attack traffic very quickly, even when the machine is newly created and has no public business value.

10. Source IP and Geolocation Analysis

Source IP addresses were extracted from failed logon events. Some source IPs were geolocated to countries including Russia, Ukraine, and China.

However, IP geolocation must be treated carefully.

Attribution Limitation

Geolocation of a source IP address does not prove the physical location or identity of the attacker. The source may represent:

- a VPN endpoint,
- a proxy server,
- a compromised host,
- cloud hosting infrastructure,
- a botnet node,
- a misattributed IP block.

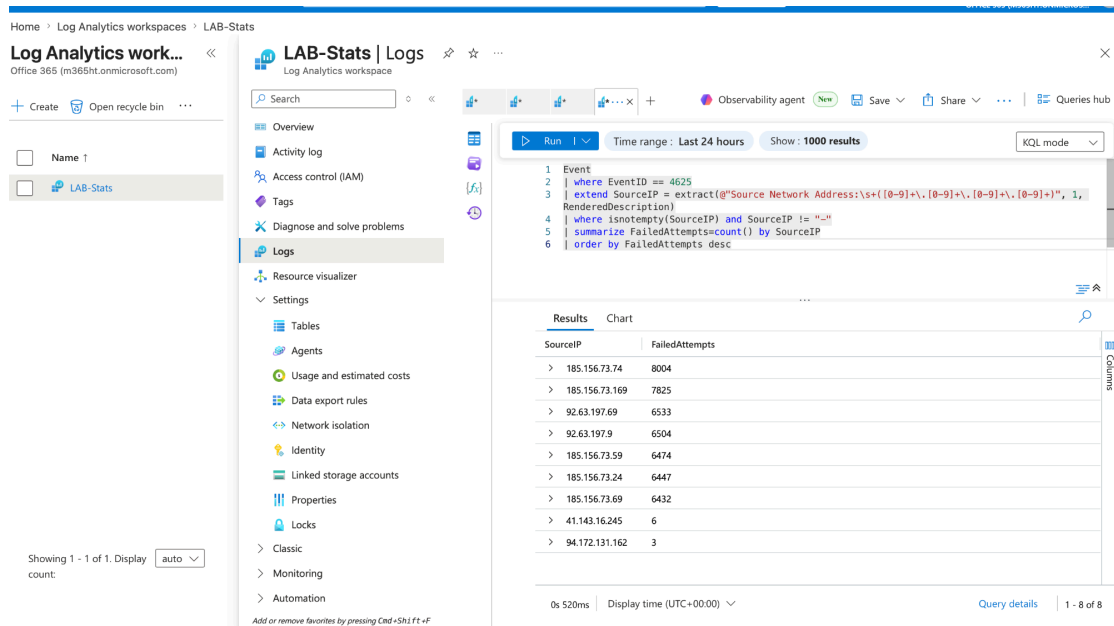


Figure 3. Captured IP addresses which failed to log in

11. Successful Logon Review

Successful logon events were reviewed to determine whether any unauthorized access occurred.

The review focused on:

- account names,
- logon type,
- source network address,
- timestamps,
- correlation with known administrator activity.

No suspicious successful logon activity was identified during the observation period.

This distinction is important because failed logons indicate attempted access, while suspicious successful logons could indicate compromise.

12. Risk Assessment

Main Risk

The main risk identified was public exposure of RDP to the internet.

Potential Impact

If weak credentials were used, an attacker could potentially:

- gain unauthorized access to the VM,
- create new accounts,
- install persistence mechanisms,
- use the VM for further attacks,
- access sensitive data if present,
- increase Azure costs through malicious activity.

Risk Level

Risk	Severity
Public RDP exposure	High
Failed brute-force attempts	Medium to High
Unauthorized successful logon	Not observed
Data exposure	Not applicable in this lab

13. Mitigation Actions

After the observation window, the public RDP exposure was mitigated.

Recommended Mitigations

Mitigation	Description
Restrict RDP source IP	Allow RDP only from a trusted public IP
Remove public RDP access	Disable inbound RDP from the internet
Use Azure Bastion	Access VM without exposing RDP publicly
Enforce strong passwords	Reduce brute-force success probability
Enable MFA where applicable	Add identity protection layer

Monitor failed logons

Alert on repeated authentication failures

Apply least privilege

Avoid unnecessary admin exposure

Keep VM patched

Reduce exploitation risk

Applied Mitigation

After completing the investigation, the inbound RDP rule was changed from:

Source: Any

Port: 3389

Action: Allow

to one of the following:

Source: Trusted IP only

Port: 3389

Action: Allow

or:

Port: 3389

Action: Deny / Rule removed

14. Lessons Learned

This lab demonstrated that:

- public-facing RDP services are quickly discovered by automated scanners,
 - failed authentication attempts can begin within a short time after exposure,
 - Azure Log Analytics can be used effectively for basic security investigations,
 - KQL is useful for filtering and aggregating security events,
 - failed logons must be distinguished from successful logons,
 - geolocation is useful for context but not reliable for attribution,
 - mitigation should follow investigation, not be ignored.
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15. Conclusion

The Azure RDP honeypot lab successfully demonstrated how quickly an internet-facing administrative service can attract authentication attempts.

Within approximately three hours, the VM received around 30,000 failed RDP login attempts. The activity was collected through Azure Log Analytics and analyzed using KQL queries. No suspicious successful logons were identified during the review.

The project highlights the risk of exposing RDP directly to the internet and demonstrates the importance of log monitoring, access restriction, and timely mitigation.